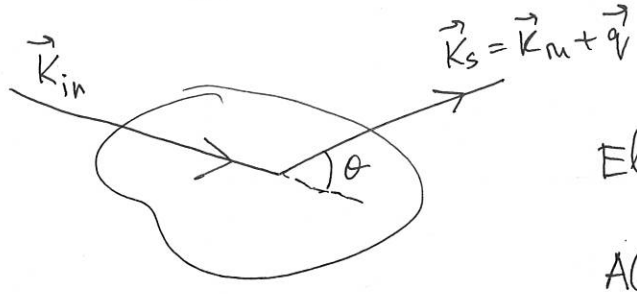
Fluctuations



Elastic scattering amplitude

$$A(\vec{q}) \sim \langle \vec{K}_s | U | \vec{K}_{in} \rangle \sim \int d\vec{x}\, e^{i\vec{q}\vec{x}} \rho(\vec{x})$$

density

Scattering intensity $S(\vec{q}) \sim \langle |A(\vec{q})|^2 \rangle \sim \langle \rho(\vec{q})^2 \rangle$

thermal average

Consider Landau-Ginzburg Hamiltonian (~~action~~ same as action for static fluctuations) for vector field $\vec{m}$

$$S = \int d\vec{x} \left(\frac{1}{2} K (\partial_\mu \vec{m})^2 + \frac{1}{2} t\, \vec{m}^2 + \frac{u}{4} \vec{m}^4 \right)$$

phase transition at $\underline{t=0}$: $\langle \vec{m} \rangle = \begin{cases} 0 & t>0 \\ \sqrt{\frac{-t}{u}}\, \hat{e}_1 , & t<0 \end{cases}$

denote $m_0 = \sqrt{\frac{-t}{u}}$

$$\vec{m} = [m_0 + \underbrace{\phi_\ell(\vec{x})}_{\text{longitudinal}}]\, \hat{e}_1 + \sum_{a=2}^{n} \underbrace{\phi_{ta}(\vec{x})\, \hat{e}_a}_{\text{transverse}}$$

n-component vector field; $\hat{e}_a$ $(a=1,2\ldots n)$ form orthonormal basis

$$(\partial_\mu \vec{m})^2 \equiv \underbrace{(\partial_x \vec{m})^2}_{\partial_x \vec{m} \cdot \partial_x \vec{m}} + (\partial_y \vec{m})^2 + (\partial_z \vec{m})^2 = \sum_{\mu,a} \partial_\mu m_a \partial_\mu m_a =$$

$$= \sum_\mu \left\{ (\partial_\mu \phi_\ell)^2 + \sum_{a=2}^{n} (\partial_\mu \phi_{t,a})^2 \right\}$$

$$\vec{m}^2 = (m_0 + \phi_\ell)^2 + \sum_a \phi_{t,a}^2 = m_0^2 + 2 m_0 \phi_\ell + \phi_\ell^2 + \vec{\phi}_t^2 .$$

$$\vec{m}^4 = \left(\vec{m}^2\right)^2 = \left(m_0^2 + 2m_0\phi_\ell + \phi_\ell^2 + \vec{\phi}_t^{\,2}\right)^2 =$$

$$= m_0^4 + 4m_0^2\phi_\ell^2 + 4m_0^3\phi_\ell + \left(\phi_\ell^2 + \vec{\phi}_t^{\,2}\right)^2 + 2\left(m_0^2 + 2m_0\phi_\ell\right)\cdot \left(\phi_\ell^2 + \vec{\phi}_t^{\,2}\right)$$

$$= m_0^4 + 4m_0^3\phi_\ell + 6m_0^2\phi_\ell^2 + 2m_0^2\vec{\phi}_t^{\,2} + O(\phi^3)$$

$$S = \int d\vec{x}\left\{\frac{1}{2}K\left(\partial_\mu\phi_\ell\right)^2 + \frac{1}{2}K\left(\partial_\mu\vec{\phi}_t\right)^2 + \frac{1}{2}t\left(\underline{m_0^2} + 2m_0\underline{\underline{\phi_\ell}} + \phi_\ell^2 + \vec{\phi}_t^{\,2}\right)\right.$$

$$\left. + \frac{1}{4}u\left(\underline{m_0^4} + 4m_0^3\underline{\underline{\phi_\ell}} + 2m_0^2\vec{\phi}_t^{\,2} + 6m_0^2\phi_\ell^2\right)\right\} = \left(t + um_0^2 = 0\right)$$

$$= V\left(\frac{1}{2}tm_0^2 + \frac{1}{4}um_0^4\right) +$$

$$+ \int d\vec{x}\left[\frac{1}{2}K\left(\partial_\mu\phi_\ell\right)^2 + \left(\frac{1}{2}t + \frac{6}{4}um_0^2\right)\phi_\ell^2\right]$$

$$+ \int d\vec{x}\left[\frac{1}{2}K\left(\partial_\mu\vec{\phi}_t\right)^2 + \left(\frac{1}{2}t + \frac{2}{4}um_0^2\right)\vec{\phi}_t^{\,2}\right]$$

Longitudinal and transverse fluctuations are <u>different</u> in the ordered phase

$$S_\ell = \int d\vec{x}\left[\frac{1}{2}K\left(\partial_\mu\phi_\ell\right)^2 + \frac{1}{2}K\xi_\ell^{-2}\phi_\ell^2\right]$$

$$S_{tr} = \int d\vec{x}\left[\frac{1}{2}K\left(\partial_\mu\vec{\phi}_t\right)^2 + \frac{1}{2}K\xi_{tr}^{-2}\vec{\phi}_t^{\,2}\right]$$

Correlation length $\longrightarrow$ $\dfrac{1}{\xi_\ell^2} = \left(t + 3um_0^2\right) = \begin{cases}\left(t + 3u\left(-\frac{t}{u}\right)\right) = -2t, & t<0\\ t, & t>0\end{cases}$

$$\frac{1}{\xi_{tr}^2} = t + um_0^2 = \begin{cases} t + u\left(\frac{-t}{u}\right) = 0, & t<0 \;!\\ t, & t>0\end{cases}$$

Fourier transform

$$\phi(\vec{x}) = \frac{1}{\sqrt{V}} \sum_q e^{i\vec{q}\cdot\vec{x}} \phi_{\vec{q}}$$

$$S_\ell = \sum_q \frac{1}{2} K \left(q^2 + \xi_\ell^{-2}\right) \left|\phi_{\ell,\vec{q}}\right|^2$$

$$S_{tr} = \sum_q \frac{1}{2} K \left(q^2 + \xi_{tr}^{-2}\right) \left|\vec{\phi}_{tr,\vec{q}}\right|^2$$

$$Z = e^{-\beta F} = \exp - V\left(\frac{1}{2} t m_0^2 + \frac{1}{4} u m_0^4\right) \cdot$$

$$\cdot \int D\phi_{\ell,\vec{q}} \; \exp - \sum_q \frac{K}{2} \left(q^2 + \xi_\ell^{-2}\right) \phi_{\ell,\vec{q}} \, \phi_{\ell,-\vec{q}}$$

$$\cdot \int D\phi_{tr,a,\vec{q}} \; \exp - \sum_q \sum_{a=2}^{n} \frac{K}{2} \left(q^2 + \xi_t^{-2}\right) \phi_{t,a,\vec{q}} \, \phi_{t,a,-\vec{q}}$$

Calculate correlation function

$$\langle \phi_{\alpha,\vec{q}} \, \phi_{\beta,\vec{q}'} \rangle = \frac{\delta_{\alpha,\beta} \, \delta_{\vec{q},-\vec{q}'}}{K\left(q^2 + \xi_\alpha^{-2}\right)}$$

Indeed

$$\langle \phi_{\alpha,\vec{q}} \, \phi_{\beta,\vec{q}'} \rangle = \frac{\int D\phi_k \left(\delta_{\alpha\beta}\right) \phi_{\vec{q}} \, \phi_{\vec{q}'} \, e^{-\sum_k \frac{K}{2}\left(k^2+\xi^{-2}\right)\phi_k \phi_{-k}}}{\int D\phi_k \, e^{-\sum_k \frac{K}{2}\left(k^2+\xi^{-2}\right)\phi_k \phi_{-k}}}$$

$$= \frac{1}{Z} \int D\phi_k \, \frac{\partial}{\partial\lambda_{\vec{q}}} \frac{\partial}{\partial\lambda_{\vec{q}'}} \exp - \sum_k \left(\frac{1}{2} K \left(k^2+\xi^{-2}\right)|\phi_k|^2 - \lambda_k \phi_k\right) \Big|_{\lambda=0}$$

$$= \frac{1}{Z} \frac{\partial^2}{\partial\lambda_q \partial\lambda_{q'}} \int D\phi_k \prod_k e^{-\frac{1}{2} K \left(k^2+\xi^{-2}\right)|\phi_k|^2 - \lambda_k \phi_k}$$

Now, complete square in the exponential:

$$-\frac{1}{2} K \left(k^2+\xi^{-2}\right) \left[\phi_k \phi_{-k} + \frac{2\lambda_k \phi_k}{K\left(k^2+\xi^{-2}\right)}\right] =$$

$$= -\frac{1}{2} K\left(k^2+\xi^{-2}\right)\left\{\left(\phi_k + \frac{\lambda_k}{K(k^2+\xi^{-2})}\right)\left(\phi_{-k} + \frac{\lambda_k}{K(k^2+\xi^{-2})}\right) - \right.$$

$$\left. - \frac{\lambda_k \lambda_{-k}}{K^2(k^2+\xi^{-2})^2}\right\}$$

Thus, linear shift $\phi_k = \tilde{\phi}_k - \frac{\lambda_{-k}}{K(k^2+\xi^{-2})}$ results in

$$\frac{1}{Z}\frac{\partial^2}{\partial\lambda_q \partial\lambda_{q'}} \int D\tilde{\phi}_k \exp + \left[\sum_k -\frac{1}{2} K \tilde{\phi}_k \tilde{\phi}_{-k} + \frac{1}{2}\frac{\lambda_k \lambda_{-k}}{K(k^2+\xi^{-2})}\right]$$

$$= \frac{\partial^2}{\partial\lambda_q \partial\lambda_{q'}} \prod_{k>0} \exp\left(\frac{1}{K(k^2+\xi^{-2})}\lambda_k \lambda_{-k}\right)$$

Here we used $\sum_k \frac{1}{2K(k^2+\xi^{-2})}\lambda_k\lambda_{-k} = \sum_{k>0}\frac{1}{K(k^2+\xi^{-2})}\lambda_k\lambda_{-k}$

$$\Rightarrow \langle \phi_{\vec{q}} \phi_{\vec{q}'} \rangle = \frac{\partial}{\partial\lambda_{q'}} \frac{\lambda_{-q}}{K(q^2+\xi^{-2})} \prod_{k>0} e^{\frac{\lambda_k\lambda_{-k}}{K(k^2+\xi^{-2})}}\Big|_{\lambda=0} =$$

(choose $q>0$ here)

$$= \frac{\delta_{q',-q}}{K(q^2+\xi^{-2})}$$

Hence $\langle \phi_{\alpha,\vec{q}} \phi_{\beta,\vec{q}'} \rangle = \frac{\delta_{\alpha\beta}\,\delta_{\vec{q},-\vec{q}'}}{K(q^2+\xi^{-2})}$ is diagonal in all indices.

Two-point correlation function in real space

$$\langle \phi_\alpha(\vec{x}) \rangle = \langle m_\alpha(\vec{x}) - \langle m_\alpha \rangle \rangle = 0$$ because action is quadratic in fluctuations.

Connected correlation function

$$G^c_{\alpha,\beta}(\vec{x},\vec{x}') = \langle (m_\alpha(\vec{x}) - \langle m_\alpha \rangle)(m_\beta(\vec{x}') - \langle m_\beta \rangle) \rangle =$$

$$= \langle \phi_\alpha(\vec{x})\, \phi_\beta(\vec{x}') \rangle = \frac{1}{V} \sum_{\vec{q},\vec{q}'} e^{i\vec{q}\cdot\vec{x} + i\vec{q}'\cdot\vec{x}'} \langle \phi_{\alpha,\vec{q}}\, \phi_{\beta,\vec{q}'} \rangle$$

$$= \frac{1}{V} \sum_{\vec{q}} \delta_{\alpha,\beta} \frac{e^{i\vec{q}(\vec{x}-\vec{x}')}}{K(q^2+\xi^{-2})} = \frac{\delta_{\alpha,\beta}}{K} \int \frac{d\vec{q}}{(2\pi)^d} \frac{e^{i\vec{q}(\vec{x}-\vec{x}')}}{q^2+\xi^{-2}}$$

<u>$d=3$</u> $$I_3 = \int \frac{d^3\vec{q}}{(2\pi)^3} \frac{e^{i\vec{q}\cdot\vec{r}}}{q^2+\xi^{-2}}$$

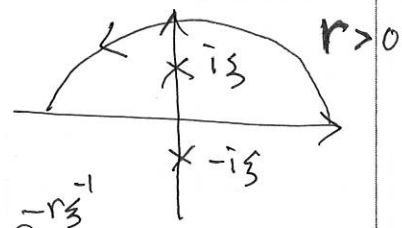
$$I_3 = \int_0^\infty \frac{dq\, q^2}{(2\pi)^3} \int_0^{2\pi} d\varphi \int_0^\pi d\theta \sin\theta \frac{e^{iqr\cos\theta}}{q^2+\xi^{-2}} =$$

$$(\eta = \cos\theta) \quad = \int \frac{dq\, q^2}{(2\pi)^2} \int_{-1}^{1} d\eta \frac{e^{iqr\eta}}{q^2+\xi^{-2}} = \int_0^\infty \frac{dq\, q^2}{(2\pi)^2} \frac{e^{iqr\eta}\Big|_{-1}^{1}}{iqr} \frac{1}{q^2+\xi^{-2}} =$$

$$= \frac{2}{(2\pi)^2 r} \int_0^\infty dq \frac{q \sin qr}{q^2+\xi^{-2}}$$

Consider now $$J = \int_0^\infty dq \frac{\cos qr}{q^2+\xi^{-2}} = \frac{1}{2} \int_{-\infty}^{\infty} dq \frac{\cos qr}{q^2+\xi^{-2}} =$$

$$= \frac{1}{2} \mathrm{Re} \int_{-\infty}^{\infty} dq \frac{e^{iqr}}{q^2+\xi^{-2}}$$



$$= \frac{1}{2} \mathrm{Re} \left\{ 2\pi i \frac{e^{iq(i\xi^{-1})}}{2i\xi^{-1}} \right\} = \frac{\pi}{2\xi} e^{-r\xi^{-1}}$$

Now observe that

$$\int_0^\infty dq\, q\, \frac{\sin qr}{q^2+\xi^{-2}} = -\frac{d}{dr}\int_0^\infty dq\, \frac{\cos qr}{q^2+\xi^{-2}} =$$

$$= -\frac{d}{dr}\left(\frac{\pi}{2\xi^{-1}} e^{-r\xi^{-1}}\right) = \frac{\pi}{2} e^{-r\xi^{-1}} = \frac{\pi}{2} e^{-r/\xi}$$

Thus, $I_3 = \frac{2}{(2\pi)^2 r} \frac{\pi}{2} e^{-r/\xi} = \frac{1}{4\pi r} e^{-r/\xi}$.

$$G^c_{\alpha,\beta}(\vec{x},\vec{x}') = \frac{\delta_{\alpha,\beta}}{K} \frac{e^{-\frac{|\vec{x}-\vec{x}'|}{\xi}}}{4\pi|\vec{x}-\vec{x}'|}$$

Thus, <u>above</u> phase transition: <u>$t>0$</u>

$$\frac{1}{\xi_{tr}} = \frac{1}{\xi_\ell} = \sqrt{t}$$

correlations of fields ϕ_ℓ, $\vec{\phi}_t$ decay exponentially

$$\xi^{-1} = \sqrt{t} \sim \sqrt{\frac{T-T_c}{T_c}}$$

<u>Below</u> phase transition, <u>$t<0$</u>

$$\frac{1}{\xi_{tr}} = 0 \quad , \text{while } \frac{1}{\xi_\ell} = \sqrt{-2t}$$

Transverse correlations decay as <u>power law</u>

$$G^c_t(\vec{r}) = \frac{\delta_{\alpha,\beta}}{4\pi K r}$$

More generally, in $\underline{d}$ dimensions

$$I_d = \int \frac{d^d \vec{q}}{(2\pi)^d} \frac{e^{i\vec{q}\cdot\vec{r}}}{q^2+\xi^{-2}} \sim \begin{cases} \dfrac{r^{2-d}}{(d-2)\, S_d} & (r \ll \xi) \\[2ex] \dfrac{\xi^{\frac{3-d}{2}}}{(d-2)\, S_d\, r^{\frac{d-1}{2}}}\, e^{-r/\xi}, & (r \gg \xi) \end{cases}$$

S_d = area of d-dim. sphere.

<u>Susceptibility</u>

$$\chi \sim \int d\vec{r}\, G^c(\vec{r})$$

Longitudinal $\quad \chi_\ell \sim \int d^3\vec{r}\, \dfrac{e^{-r/\xi}}{4\pi r} \sim \int_0^{\xi_\ell} \dfrac{dr\, r^2}{r} \sim \xi_\ell^2$

Transverse $\quad \chi_{tr} \sim \int d^3 r\, \dfrac{1}{4\pi r} \sim \int_0^L \dfrac{dr\, r^2}{r} \sim \underline{L^2 \gg \xi_\ell^2}.$

L = system size

At the critical point, $\underline{t=0}$, $\xi_\ell \sim \dfrac{1}{\sqrt{t}} = \infty$

$\xi_{tr} \sim \dfrac{1}{\sqrt{t}} = \infty$

$\Rightarrow$ both susceptibilities diverge there!

Fluctuations corrections

$$Z = \exp\left[-V\left(\frac{1}{2}t m_0^2 + \frac{1}{4}u m_0^4\right)\right] \cdot$$

$$\cdot \int D\phi_{l,\vec{q}} \exp -\sum_q \frac{K}{2}\left(q^2+\xi_l^{-2}\right)\phi_{l,\vec{q}}\,\phi_{l,-\vec{q}}$$

$$\cdot \int D\phi_{tr,\vec{q}} \exp -\sum_q \frac{K}{2}\left(q^2+\xi_{tr}^{-2}\right)\phi_{tr,\vec{q}}\,\phi_{tr,-\vec{q}}$$

Consider $\int d\varphi_{\vec{q}}\; e^{-\frac{1}{2}K(q^2+\xi^{-2})|\varphi_{\vec{q}}|^2} = \dfrac{2\pi}{\left(K(q^2+\xi^{-2})\right)^{1/2}} =$

$$= (2\pi)\, e^{-\frac{1}{2}\ln\left[K(q^2+\xi^{-2})\right]}$$

Hence $\int D\phi(\vec{q}) \exp -\frac{1}{2}\sum_q K(\vec{q}^2+\xi^{-2})|\phi(\vec{q})|^2 =$

$$= \text{cust} \cdot \exp -\frac{1}{2}\sum_{\vec{q}} \ln\left[K(\vec{q}^2+\xi^{-2})\right] =$$

$$= \exp -\frac{V}{2}\int \frac{d\vec{q}}{(2\pi)^d} \ln\left[K(\vec{q}^2+\xi^{-2})\right].$$

$$Z = e^{-\beta F}, \quad \beta F = -\ln Z$$

$$\beta F = V\left(\frac{1}{2}t m_0^2 + \frac{1}{4}u m_0^4\right) + \frac{V}{2}\int \frac{d\vec{q}}{(2\pi)^d} \ln K(q^2+\xi_l^{-2})$$ from longitudinal

$$+ V\,\frac{n-1}{2}\int \frac{d\vec{q}}{(2\pi)^d} \ln K(q^2+\xi_t^{-2})$$

from $(n-1)$ transverse modes.

Free energy density $f = \dfrac{F}{V}$

Now consider fluctuations:

$\underline{t>0}$ $\quad \xi_\ell^{-2} = \xi_t^{-2} = t$

$$\Rightarrow \beta f_{fluct} = \frac{1+n-1}{2} \int \frac{d\vec{q}}{(2\pi)^d} \ln\left[K\vec{q}^2 + t)\right]$$

$$-\frac{\partial^2(\beta f)}{\partial t^2} = \frac{n}{2} \int \frac{d\vec{q}}{(2\pi)^d} \frac{1}{(K\vec{q}^2+t)^2}$$

The integral $J_d = \int \frac{d^d\vec{q}}{(2\pi)^d} \frac{1}{(q^2+\xi^{-2})^2} \sim \int_0^\Lambda dq \frac{q^{d-1}}{(q^2+\xi^{-2})^2}$

<u>For $d>4$</u>, the integral diverges at the upper cut-off ($\Lambda \sim \frac{1}{a_0}$, a_0 = lattice spacing) ($\Lambda \gg \xi^{-1}$)

$$\int^\Lambda dq\, q^{d-5} \sim (\Lambda)^{d-4} \sim (a_0)^{4-d}$$

<u>For $d<4$</u>, J_d converges at $\Lambda \Rightarrow$ can set $\Lambda = \infty$

$$J_{d<4} \approx \int_0^\infty dq \frac{q^{d-1}}{(q^2+\xi^{-2})^2} = \left(q = \frac{x}{\xi}\right) =$$

$$= \int_0^\infty \frac{dx}{\xi} \frac{x^{d-1}\,\xi^{-(d-1)}}{(x^2+1)^2\,\xi^{-4}} = \xi^{4-d} \underbrace{\int_0^\infty dx \frac{x^{d-1}}{(x^2+1)^2}}_{\text{number}}$$

$$\Rightarrow J_d \sim \begin{cases} (a_0)^{4-d} & \text{for } d>4 \\ (\xi)^{4-d} & \text{for } d<4 \end{cases}$$

But $\xi \sim t^{-1/2} \Rightarrow \xi^{4-d} \sim (t)^{\frac{d-4}{2}}$ <u>diverges</u> $t \to 0$, $d<4$.

Heat capacity $C_V = T\left(\frac{\partial S}{\partial T}\right)_V = -T\left(\frac{\partial^2 F}{\partial T^2}\right)_V$

In terms of convenient combination

$$\beta f = \frac{F}{TV}$$

$$C_V = -T\left(2\,\frac{\partial(\beta f)}{\partial T} + T\,\frac{\partial^2(\beta f)}{\partial T^2}\right)$$

More derivatives = stronger divergence,

$\Rightarrow$ focus on singular part $C_{sing} = -T^2\,\frac{\partial^2(\beta f)}{\partial T^2}$

In terms of reduced temperature t:

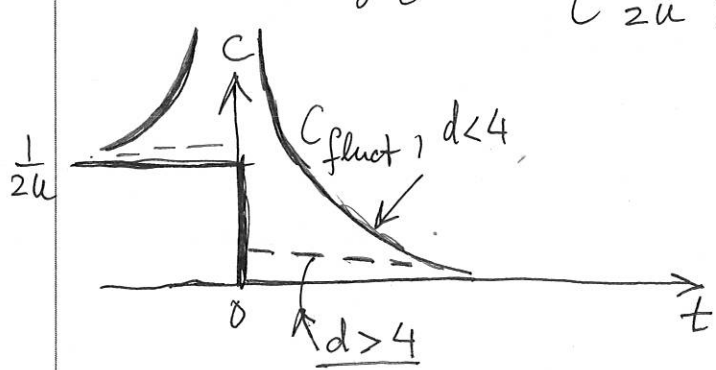
$$C_{sing} \sim -\frac{\partial^2(\beta f)}{\partial t^2}$$

First, analyze $\beta f_0 = \frac{1}{2} t m_0^2 + \frac{1}{4} u m_0^4$

$$\left(m_0^2 = -\frac{t}{u} \text{ for } t<0\right)$$

$$\Rightarrow \beta f_0 = \begin{cases} 0, & t>0 \\ \frac{1}{2}t\left(\frac{-t}{u}\right) + \frac{1}{4}u\left(-\frac{t}{u}\right)^2 = -\frac{t^2}{2u} + \frac{t^2}{4u} = -\frac{t^2}{4u}, & \underline{t<0} \end{cases}$$

$$\Rightarrow -\frac{\partial^2(\beta f_0)}{\partial t^2} = \begin{cases} 0, & t>0 \\ \frac{1}{2u}, & t<0 \end{cases}$$



Observe that

$$-\frac{\partial(\beta f)}{\partial t} = \frac{t}{2u} = 0$$

at $t=0$, i.e. it is <u>not singular</u>

$$C = C_0 + C_{fluct} \sim \begin{cases} C_0 \ , \text{ for } d>4 \\ C_0 + \left(t^{\frac{d-4}{2}}\right) \ , \ d<4 \end{cases}$$

fluctuations produce t-independent correction

fluctuational correction $\sim t^{\frac{d-4}{2}}$ is more important than original C_0

Thus, saddle-point analysis, which relies on minimization of $S = \int d\vec{x} \ \frac{1}{2}K(\partial_\mu \vec{m})^2 + \frac{1}{2}t\,\vec{m}^2 + \frac{1}{4}u\,\vec{m}^4$ ~~simply does~~ does not capture physics of strong fluctuations near the critical point.

<u>$d=4$</u> upper critical dimension
(depends on symmetry of the action S only)

$$\begin{cases} d>4: \text{ fluctuations not important} \\ d<4: \quad \text{— — — —} \ \underline{\text{are}} \text{ important.} \end{cases}$$

<u>Ginzburg criterion, $d<4$:</u>

$$\xi(T) = \frac{\xi_0}{|t|^{-1/2}} \quad \left(t = \frac{T-T_c}{T_c}\right)$$

$$\left.\begin{array}{l} C_{fluct} \sim C_0 = \frac{1}{2u} \\ \downarrow \\ |t|^{\frac{d-4}{2}} \end{array}\right\}$$

sufficiently close to the transition $C_{fluct} > C_0$. This defines temperature t_G where simple LG analysis fails to describe critical fluctuations.

$$|t_G| \sim C_0^{\frac{2}{d-4}} = \left(\frac{1}{C_0}\right)^{\frac{2}{4-d}}.$$

$d=4$ upper critical dimension where fluctuations invalidate description of the critical point.

$d=2$ lower critical dimension where fluctuations destroy the ordered phase altogether (at $T>0$).